



University of Vavuniya

Second Examination in Information Technology - 2022

First Semester - January/February 2024

IT2153 Computer Graphics (Theory)

Answer any Four Questions Only

Time Allowed: Two Hours

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1. (a) Compare and contrast *raster graphics* and *vector graphics*. [20%]  
(b) Briefly describe what is meant by *scan conversion* in Computer Graphics. [10%]  
(c) Write *DDA* line drawing algorithms to draw lines when  $|m| \leq 1$ . [20%]  
(d) Derive the necessary equations for the *Bresenham* line drawing algorithm for a positive slope less than one and use it to determine the coordinate points of a line between (10,20) and (18,25). [50%]  
*check again*
  2. (a) Describe how the symmetrical property of a circle can be used to *scan converting* a circle with the aid of a diagram. [10%]  
(b) Derive the necessary equations to draw a circle using the *Midpoint* circle drawing algorithm. [20%]

[This question is continued on the next page...]

- (c) Determine the pixel positions along the circle octant of radius  $r = 10$  from  $x = 0$  to  $x \leq y$  centered at the origin using the algorithm in part 2(b). [30%]
- (d) Write procedures for each of the following 8-connected filling algorithms:
- Boundary fill
  - Flood fill
- [40%]
3. (a) Derive equations for *window to viewport* transformation of a point  $(w_x, w_y)$  in the window maps to the point  $(v_x, v_y)$  in the viewport in such a way as to preserve proportion. Clearly state all assumptions made. [30%]
- (b) i. Describe the *Cohen-Sutherland* clipping algorithm with the criteria to trivially accept or reject a line against a given clipping window. [20%]
- ii. Give the necessary equations to determine the coordinates of the intersection points of a line with the clipping window boundaries  $x_{min}$ ,  $x_{max}$ ,  $y_{min}$  and  $y_{max}$ . [20%]
- iii. Use the *Cohen-Sutherland* line clipping algorithm to clip a line P1(70,20) and P2(100,10) against a window with the coordinates at the lower left hand corner of (50,10) and the upper right hand corner of (80,40). [30%]
4. (a) Describe the characteristics of *parallel* and *perspective* projections in three dimensional graphics with the aid of a diagram. [20%]
- (b) Compare and contrast *orthographic* and *oblique* projections. [20%]
- (c) Describe what is meant by *vanishing point* in Computer Graphics. [10%]
- (d) Derive a homogeneous transformation matrix for perspective projections of a point  $P(x, y, z)$  on the  $XY$  view plane, if the centre of the projection is  $(0,0,-d)$ . [20%]
- (e) Suppose that the coordinates of an object ABCD in three dimensional space are A(-15,15,20), B(10,15,20), C(25,0,20), and D(0,0,20). Find the coordinates of the projected image on the  $XY$  view plane using the projection method in part 4(d) if the centre of the projection is  $(0,0,-20)$  and sketch the object. [30%]

5. (a) Consider the four control points  $P_0(1,0)$ ,  $P_1(3,3)$ ,  $P_2(6,3)$ , and  $P_3(8,1)$  of a cubic Bézier curve.

Determine any three points except the first and last points that lie on the curve. [15%]

- (b) i. Find the coordinate functions  $x(t)$ ,  $y(t)$ , and  $z(t)$  of the cubic Bézier curve with the control points  $P_0(0,0,0)$ ,  $P_1(1,0,0)$ ,  $P_2(0,1,0)$ , and  $P_3(0,0,1)$ . [30%]

- ii. Compute the tangents at the end points of the cubic Bézier curve in part b(i). [20%]

- (c) i. Derive the equation for quadratic Bézier curve using De Casteljau algorithm. [15%]

- ii. Determine the algebraic expression of the quadratic Bézier curve with the control points  $P_0(0,0)$ ,  $P_1(2,3)$ , and  $P_2(5,-1)$ . [20%]

$$x(t) = (1-t)^3 + 9t(1-t)^2 + 18t^2(1-t) + 8t^3$$

$$y(t) = 9t(1-t)^2 + 9t^2(1-t) + t^3$$